

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 26

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Use a calibrated sound meter (dB(A)) at fixed distances (eg 10, 50, 100, 250, 500 m) from the track behind each barrier and at a control site with no barrier • Standardise the noise source: same cars / same race or a standard test noise; same time of day, wind speed / direction, weather; same height of meter above ground • Take repeated readings at each distance and calculate means (and standard deviation) to assess variability • Use a measure such as LA10 or Leq over a set period rather than a single peak reading • Compare reduction in dB relative to the control at each distance; plot noise level against distance for each barrier; statistical test (t-test) to assess significance	5
01.2	• Wind carries sound / affects how far it travels so different wind directions would change readings independently of the barrier	1
01.3	• LA10 is the noise level exceeded for 10% of the period - a measure of the noisiest periods • LA90 is the level exceeded for 90% of the period - a measure of the quietest / background level	2
01.4	<i>Any 2 from:</i> • Double / triple glazing • Acoustic insulation in walls / roof • Bedrooms on the side facing away from the track • Dense hedges / trees and earth mounds in gardens	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 293-295, 418; AQA spec 3.4.3.2.13.	
02.1	• B	1

Question	Marking guidance	Marks
02.2	• Number all possible sampling sites (eg on a grid map) • Select the sites using a random number generator / table; (regular shape: grid coordinates from random numbers)	2
02.3	• Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis • So the sample does not represent the whole area and conclusions are invalid	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
03.1	• Oxygen	1
03.2	• A change to one process causes other processes to alter as a direct result (chains of linked processes / feedback) • There is still a lot that is not understood about how atmospheric processes work	2
03.3	• The surface warmed in the tropics heats the air above; warm winds (south-westerlies from the Caribbean) carry heat energy to higher latitudes • Winds carry water vapour (evaporated from warm oceans) to areas that would otherwise receive little precipitation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-88; AQA spec 3.2.1.1.	
04.1	• A fluid in a buried ground loop absorbs low-temperature heat from the ground; at the expansion valve the fluid boils at low pressure absorbing this heat • A compressor raises the pressure so the gas condenses, releasing heat at a higher temperature inside the building • The electricity only moves heat rather than creating it, so up to four times as much heat is delivered as electricity used (less if the source is colder)	3
04.2	• Parabolic reflectors focus sunlight onto tubes of oil, increasing the energy density and reaching high temperatures • The heat is transferred to molten salt (potassium nitrate) stored in large insulated tanks at up to 550 °C • The stored heat boils water to drive steam turbines whenever electricity is required, including at night	3
04.3	• Reflectors can only focus parallel rays of direct sunlight; cloud, dust and smoke scatter light so it cannot be concentrated • PV cells convert diffuse as well as direct light, though output is lower	2
04.4	<i>Any 2 from:</i> • Manufacture: extraction / processing of silicon, metals and plastics; toxic wastes (silicon tetrachloride, cadmium) • Land take by large solar farms competing with other uses / affecting desert flora and fauna • Water for cleaning panels in arid areas • Disposal of old panels	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 200-202; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
05.1	• Production prevention: desulfurisation of fuels before combustion; low-temperature (fluidised bed) combustion to prevent NO_x; smokeless fuels • Prevention of release: electrostatic precipitators / bag filters / cyclones for smoke; catalytic converters; flue gas desulfurisation; bund walls; landfill liners • Post-release remediation: oil spill clean-up (booms, skimmers, dispersants, bioremediation); phytoremediation of metal-contaminated land; liming of acidified lakes • Alternative processes: electric vehicles instead of petrol / diesel; renewable energy instead of fossil fuels; pyrethroids instead of organochlorines; fuel cells instead of internal combustion engines	4
05.2	• Remediation is often incomplete / cannot recover all the pollutant (eg oil spills, dispersed pollutants), so some damage still occurs • Prevention avoids the damage entirely and is usually cheaper than clean-up / avoids polluter-pays costs; moving from post-production treatment to non-release is a circular economy principle	2
05.3	• Satellites detect and track the extent / movement of oil slicks over large areas, quickly and repeatedly, so clean-up can be directed and polluters identified • GPS tracking monitors ship positions so navigation errors / collisions / groundings are reduced and offshore shipping routes are enforced • Polymers with a high surface area adsorb metal ions from contaminated water / effluent so they can be removed and recovered • Plants that absorb heavy metals (hyperaccumulators) are grown on contaminated land, harvested and disposed of / metals recovered, cleaning the soil	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
06.1	• The water was recharged during or before the last ice age (10 000-20 000 years ago) when the climate was wetter • Current recharge is little or none so any abstraction reduces the stored volume • The large volume means shortages take a long time to appear, but eventually the supply runs out; ancient water is often saline, causing soil salinisation	3
06.2	• Two satellites whose orbits are affected by the force of gravity • The mass of water in aquifers below changes gravity, so changes in the orbit / distance between the satellites indicate changes in groundwater mass • Advantage: the whole Earth is surveyed every 30 days, whereas borehole water-table measurements are slow and hard to coordinate over large areas	3
06.3	<i>Any 3 from:</i> • Low-pressure irrigation pipes to reduce leaks • Growing crops that need less water than wheat / maize • Charging consumers by volume used (metering) • Inter-basin transfer of water 450 km from the Yangtze	3
06.4	• Trees intercept rainfall and transpire large volumes, reducing soil moisture and the water percolating down to the aquifer	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
07.1	• In general, higher GDP per capita is associated with higher energy use per capita (positive correlation) • But not proportional: B has a high GDP but much lower energy use than A or E; C uses relatively high energy for its GDP	2
07.2	• Affluent people buy more consumer goods, appliances and travel more / drive larger, less efficient vehicles and heat larger homes • Manufacture and transport of the extra products bought uses energy (indirect use); larger houses use more energy to build • Affluent countries can afford energy for non-essential uses; cheap local energy (eg petrol in the USA) encourages wasteful use	3
07.3	<i>Any 2 from:</i> • Higher energy prices / taxes encouraging conservation • Culture of energy conservation (eg Scandinavia after past shortages) / environmental awareness • Milder climate needing less heating or air conditioning • Service-based economy: heavy industry (steel, aluminium) has moved to other countries so energy embodied in imports is not counted • Smaller / more efficient vehicles, better public transport, better insulated buildings	2
07.4	• Decline of heavy industries such as steel and aluminium smelting in the UK • The energy-intensive goods are now manufactured abroad (eg China) so the energy is counted in the exporting country; also efficiency improvements	2
07.5	• Water: pumping, treatment / sterilisation, sewage treatment; Agriculture: machinery, fertiliser manufacture (Haber process), irrigation pumps, processing / refrigeration	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174; AQA spec 3.3.1.	
08.1	• Negative feedback: a change causes other changes that reduce the rate or size of the original change, re-establishing equilibrium • Positive feedback: a change causes other changes that increase the original change	2
08.2	• Increased low-level cloud: higher temperatures increase evaporation and condensation, producing more cloud; clouds have a high albedo so more sunlight is reflected and warming is reduced • Increased photosynthesis: higher temperatures and CO₂ increase photosynthesis, removing CO₂; if stored in woody tissue atmospheric CO₂ rises less	4

Question	Marking guidance	Marks
08.3	- Permafrost: frozen waterlogged Arctic soils contain dead organic matter that decomposed anaerobically producing methane trapped in the frozen ground - Warming thaws the permafrost, releasing methane (a powerful greenhouse gas) and allowing further decomposition, causing more warming - Methane hydrate: methane from decomposed organic matter in deep sea sediments is held as a solid under high pressure and low temperature - If deep water warms the hydrate melts, releasing methane into the atmosphere and causing further warming	4
08.4	<i>Any 3 from:</i> - Reduced albedo as snow / ice melts - Ocean acidification reducing coral survival and carbon sequestration - Increased forest and peat fires releasing CO₂ - Faster decomposition of soil organic matter releasing CO₂ - Increased water vapour (a greenhouse gas) in warmer air - Increased cirrus (high) cloud which traps infrared	3
08.5	- A point at which natural processes triggered by human-caused warming become self-sustaining / unstoppable - So that stopping the original human activity would no longer stop climate change (eg runaway permafrost melt, methane hydrate release, forest fires)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-104, 371-372; AQA spec 3.2.1.2, 3.6.1.	
09.1	<i>Any 4 from:</i> - Topography: a narrow exit from a large deep basin so a small dam holds a large volume - Geology: impermeable rock beneath (no seepage), strong enough to support the dam, no faults / seismic activity - Catchment area: large enough with regular, reliable rainfall / river inflow; climate not too hot (evaporation losses) - Catchment land use: low pollution risk (industry, pesticides); low soil erosion (sedimentation reduces capacity) - Infrastructure / proximity to demand: access for construction, cost of transporting water - Existing land use: value of what is lost (urban areas, conservation sites, farmland) - loss:benefit analysis	4
09.2	- Advantage: fewer land-use conflicts as no land is flooded; large volume near coastal cities - Disadvantages: destroys / changes important intertidal habitats (mudflats for wading birds); obstacle to shipping; collects pollution from the whole river catchment; sedimentation	3
09.3	- Very energy-intensive and expensive (reverse osmosis / distillation) - Only worthwhile where seawater is available and freshwater supplies are inadequate (eg Malta, Middle East)	2
09.4	Surplus water in wet periods (when infiltration capacity is exceeded and runoff would go to the sea) is pumped underground or diverted into lagoons to infiltrate, so abstraction in dry periods is balanced by recharge	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-124; AQA spec 3.2.2.5.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • State of matter: solids deposit near the source, gases disperse widely in the atmosphere; density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel • Persistence / half-life: persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution; non-persistent pollutants (sewage, organophosphates) can be allowed to degrade • Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk; LD50 / MDAF / MDNF used to set doses • Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) • Reactivity: reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants • Adsorption onto clay / sediments immobilises metals and caesium; can be used deliberately (activated carbon filters, polymer adsorption) • Mutagenic / carcinogenic / teratogenic action means no safe dose; exposure kept As Low As Reasonably Achievable (ALARA) • Synergism (ozone + SO₂; smoke + SO₂) means pollutants must be considered together • Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring • Evaluation: properties of new chemicals can be predicted from similar ones (precautionary principle); understanding allows prevention rather than clean-up	9
10.2	• The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	

Question	Marking guidance	Marks
11.1	Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Historical data limitations; proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • Satellite sensors: visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring; passing through cloud • Sensor carriers: ROVs / drones, AUVs, balloons, aircraft, animals; data recorders for abiotic factors; GPS tracking of floats and sampling sites; Argo floats and ocean monitoring • Climate models and their assumptions; monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • Evaluation: improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	25

Question	Marking guidance	Marks
11.2	Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why non-renewable: formed too slowly; geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock; Lasky's principle; cut-off ore grade and its drivers • Extraction technologies: mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • Reducing demand: recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • Environmental limits: impacts of mining and their control; energy costs rising with lower grades; conflicts • Judgement: technology can extend supplies substantially but only within energy and environmental constraints; circular economy essential Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	25